ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ-3 **SUMMER SEMESTER, 2022-2023**

DURATION: 30 MINUTES **FULL MARKS: 15**

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all** **4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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1. In Figure 1, the area of the shaded regions A and B, portrayed beneath the two functions $f_a(x)$ and $f_b(x)$ respectively, are equal. The aforementioned functions are, 5 (CO2) (PO1)

$$f_a(x) = -x^2 + k$$
$$f_b(x) = x^3 + x^2$$

Using the Composite Trapezoidal Rule, with $n = 8$ segments, approximate the value of k.

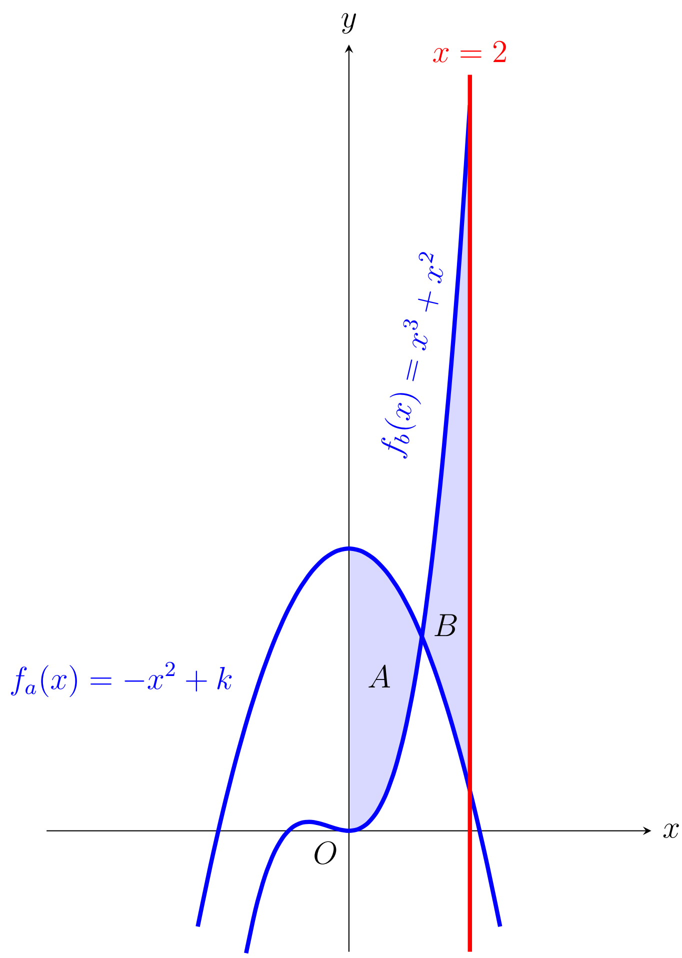

Figure 1: The shaded regions A and B.

Solution:

Since the shaded regions A and B cover an equal area, we can include the common overlapping region and posit that,

$$\int_0^2 f_a(x)dx = \int_0^2 f_b(x)dx \tag{1}$$

We can approximate the integral value of a function $f(x)$ using composite trapezoidal rule with n segments, each of width $h = \frac{b-a}{n}$, between the lower-limit a and the upper-limit b as follows,

$$\int_a^b f(x)dx = \frac{h}{2}\left[f(a) + 2\left\{\sum_{i=1}^{n-1} f(a+ih)\right\} + f(b)\right]$$

Hence, we can expand Equation 1 by applying this formula, with $n = 8$ and $h = \frac{2}{8} = 0.25$.

$$\frac{h}{2}\left[f_a(0) + 2\left\{\sum_{i=1}^{7} f_a(0+ih)\right\} + f_a(2)\right] = \frac{h}{2}\left[f_b(0) + 2\left\{\sum_{i=1}^{7} f_b(0+ih)\right\} + f_b(2)\right]$$

$$\begin{aligned}
\Rightarrow & f_a(0) + 2 \times [f_a(0.25) + f_a(0.5) + f_a(0.75) + f_a(1) + f_a(1.25) + f_a(1.5) + f_a(1.75)] + f_a(2) \\
& = f_b(0) + 2 \times [f_b(0.25) + f_b(0.5) + f_b(0.75) + f_b(1) + f_b(1.25) + f_b(1.5) + f_b(1.75)] + f_b(2) \\
\Rightarrow & k + 2 \times [-0.0625 + k - 0.25 + k - 0.5625 + k - 1 + k - 1.5625 + k - 2.25 + k - 3.0625 + k] \\
& + (-4 + k) = 0 + 2 \times [0.078125 + 0.375 + 0.984375 + 2 + 3.515625 + 5.625 + 8.421875] + 12 \\
\Rightarrow & 16k - 21.5 = 54 \\
\Rightarrow & k \approx 4.71875
\end{aligned}$$

Rubric:

- 1 point for the correct relationship between the integrals of $f_a(x)$ and $f_b(x)$.
- 1 point for the correct composite trapezoidal rule formula.
- 2 points for the correct application of the formula to expand the relationship.
- 1 point for the correct (or an approximately correct) answer.

2. Suppose, you are given a set of n data points (x_1, y_1), (x_2, y_2), $\ldots$, (x_n, y_n) and you want to regress the data to an mth order Polynomial model, where $m < n$. How would you choose a suitable value of m? Explain with proper illustration(s). 4 (CO3) (PO1)

Solution:

In order to choose a suitable value of m, we need to maintain a Bias-Variance tradeoff.

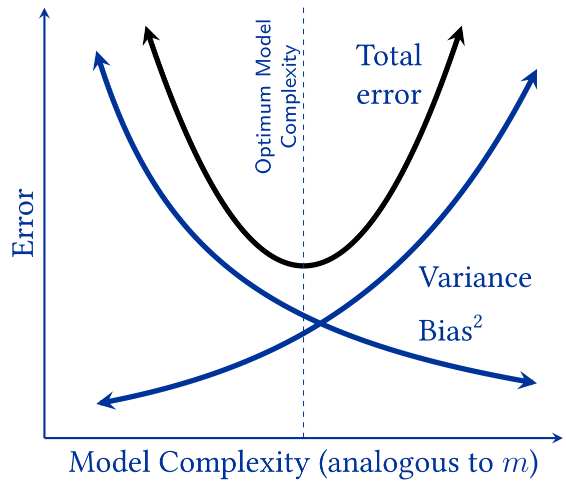

Figure 2: Bias-Variance Tradeoff

If the value of m is too high, the polynomial model will overfit the given n data points and result in high variance. On the other hand, if the value of m is too small, the polynomial model will underfit the given n data points and result in high bias.

Rubric:

- 2 points for explaining how the Bias-Variance tradeoff works.
- 2 points for nice illustration(s) related to the Bias-Variance tradeoff.

3. Suppose, you are monitoring the progress of a homogeneous chemical reaction in order to calculate the rate constant k and the order of the reaction n. The rate law expression for the reaction can be represented with the Power model 5 (CO2) (PO1)

$$-r = kC^n$$

where r is the reaction rate and C is the concentration of the reactant. The values of reaction rate $-r$ and the respective concentration C of the reactant that you have calculated over the course of time are shown in the table below —

Concentration, C $(gmolL^{-1})$	**Reaction Rate,** $-r\ (gmolL^{-1}s^{-1})$
4	0.398
2.25	0.298
1.45	0.238
1	0.198
0.65	0.158
0.25	0.098
0.006	0.048

Table 1: Chemical kinetics of the observed homogeneous reaction.

Determine the value of the rate constant k and the order of the reaction n using the data provided in Table-1. You are allowed to apply data transformation.

Solution:

Let's linearize the data at first. Taking the natural logarithm of both sides of the given rate law expression, we get,

$$\ln(-r) = \ln(k) + n\ln(C)$$

Let, $z = \ln(-r)$, $w = \ln(C)$, and $a_0 = \ln(k)$; implying that $k = e^{a_0}$ and $n = a_1$. The straight line that we end up with is

$$z = a_0 + a_1 w$$

Using the calculator, we obtain the following values for the $\hat{n} = 7$ transformed data points.

$$\sum_{i=1}^{\hat{n}} w_i = -4.3643,\ \sum_{i=1}^{\hat{n}} z_i = -12.391,\ \sum_{i=1}^{\hat{n}} w_i z_i = 16.758,\ \sum_{i=1}^{\hat{n}} w_i^2 = 30.998$$

Now, let's apply the formulae for determining the corresponding linear regression parameters a_0 and a_1.

$$a_1 = \frac{\hat{n}\sum_{i=1}^{\hat{n}} w_i z_i - \sum_{i=1}^{\hat{n}} w_i \sum_{i=1}^{\hat{n}} z_i}{\hat{n}\sum_{i=1}^{\hat{n}} w_i^2 - \left(\sum_{i=1}^{\hat{n}} w_i\right)^2} = 0.31943,\ a_0 = \frac{\sum_{i=1}^{\hat{n}} z_i}{\hat{n}} - a_1\frac{\sum_{i=1}^{\hat{n}} w_i}{\hat{n}} = -1.5711$$

Hence, we obtain

$$\begin{aligned} k &= e^{-1.5711} \\ &= 0.20782 \end{aligned}$$

$$n = 0.31941$$

Plugging these values into the given Power model equation, we get,

$$-r = 0.20782 \times C^{0.31941}$$

Rubric:

- 1 point for the correct linear representation.
- 1 point for the correct a_1 formula.
- 0.5 point for the correct a_1 value.
- 1 point for the correct a_0 formula.
- 0.5 point for the correct a_0 value.
- 0.5 point for the correct k value.
- 0.5 point for the correct n value.

4. What is Regression Analysis? Provide a precise formulation of the problem statement of a regression model. 1 (CO2) (PO1)

Solution:

In statistical modeling, regression analysis is a set of statistical processes for estimating the relationships between a dependent variable and one or more independent variables.
Formally, the problem statement for a regression model is:

> Given n data points $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, the goal is to determine the function $y = f(x)$ that is the best-fit predictor function of the underlying data distribution.

Rubric:

- 0.5 point for a correct definition.
- 0.5 point for a correct problem formulation.